Classical Wave-Optical Emulation of Quantum-Gate Algebra on a Smartphone, with a Falsification Protocol

Svetoch Series, Paper IV of VI

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Abstract

We describe a family of smartphone experiments that **emulate the linear algebra of quantum gates using classical wave optics**, and we frame them with deliberate honesty. The Svetoch platform — an unmodified phone whose OLED screen displays patterns, a ordinary flat mirror, and the front camera — has been used to implement procedures labelled "Hadamard", "QFT", "Grover", "Deutsch–Jozsa", "CNOT", "teleportation", Hong–Ou–Mandel (HOM), CHSH/Bell, and BB84. We state plainly that **none of these is quantum computation**: there are no single photons, no entanglement, no Bell-inequality violation, and no quantum no-cloning; the light from an OLED is bright, classical, and largely incoherent. What the experiments do is encode a "qubit state" as a spatial/phase pattern of brightness and realize gates as classical interference and Fourier-optical transforms, with "measurement" as $\arg\max$ over intensities — a legitimate, pedagogically valuable *classical analogue*, not a quantum device. The defensible contributions are three: (a) a lock-in / phase-sensitive optical correlator referenced to the 120 Hz display refresh — explicitly a **classical** correlator, not a Bell test; (b) true random numbers from CMOS shot noise, a real physical entropy source (with acknowledged prior art); and, above all, (c) a rigorous **falsification protocol** — the mirror-less control — that separates real optical effects from CPU artifacts. With the mirror the emulations produce structured metrics; without it they collapse to noise and coin-flips (HOM visibility $0.056 \to 0.0007$; Deutsch–Jozsa correct $\to 50\%$; channel capacity $16 \to 1$ bit). That collapse is the honesty test, and the method for performing it is this paper's real novelty.

Keywords: classical optical analogue, wave optics, Fourier optics, quantum-gate emulation, lock-in correlator, falsification control, shot-noise TRNG, smartphone optics.

1 Introduction

Quantum computing is glamorous, and the temptation to call a clever optics demo "quantum" is strong. A phone that displays a two-lobe pattern and reads back an interference fringe *looks* like a qubit; an $\arg\max$ over two intensities *looks* like a measurement collapse. The Svetoch repository contains a whole family of such demos, with names borrowed directly from quantum information science. This paper exists to draw a hard line around what they are.

Our stance is simple and we hold it throughout: **these experiments emulate the finite-dimensional linear algebra of quantum gates with classical wave optics, and nothing more.** A single qubit lives in $\mathbb{C}^2$; a Hadamard is a 2×2 unitary; the quantum Fourier transform is a DFT. Linear, unitary, finite-dimensional maps can be realized — approximately and classically — by interference and Fourier optics, because classical wave amplitudes also superpose linearly. What cannot be realized this way is everything that makes quantum computation *quantum*: genuine entanglement across a tensor product, exponential state-space scaling, Bell-inequality violation, and no-cloning. We claim none of these, and we say so loudly, because the credibility of the whole Svetoch series rests on not overclaiming.

This is Paper IV of five. Paper I establishes the OLED–mirror–camera channel as an analog matrix engine and the falsification control underpinning the series; Paper II covers device-physics ML primitives; Paper III a mirror-free thermo-optical channel; Paper V a liquid-sensing application. Here we (i) explain precisely what is and is not being computed, (ii) describe the one piece of the "quantum" cluster worth keeping — a classical lock-in correlator — on honest terms, and (iii) give the falsification protocol that distinguishes a real optical effect from CPU theatre.

2 What is and isn't being computed

2.1 Encoding a "qubit" as a classical spatial/phase pattern

A single-qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$, is two complex numbers up to a phase. In the optical analogue we encode the amplitudes as two spatially separated modes on the OLED: two screen regions whose relative brightness sets $|\alpha|^2{:}|\beta|^2$ and whose displayed relative phase (a fringe offset) sets $\arg(\beta/\alpha)$. The "state" is a classical field

$$E(x) = \alpha\, u_0(x) + \beta\, u_1(x), \tag{1}$$

a coherent (within one displayed pattern) superposition of two spatial modes. This is exactly Spreeuw's classical analogy of a qubit [1]: the Hilbert-space algebra of one or two qubits has a faithful representation in the spatial modes of a single classical beam. It is an analogy — and a useful one — but a beam carrying 2^n modes costs resources growing with 2^n, the precise sense in which it is *not* a quantum computer.

2.2 Gates as interference and Fourier optics

On this encoding the gates are ordinary wave optics:

Hadamard. $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ is a 50:50 mixing of the two modes — a beam-splitter / two-slit interference whose readout intensities realize the sum and difference of the inputs.

QFT. A discrete Fourier transform, approximated by Fraunhofer/Fresnel diffraction (Section 2.3). Fourier-optical DFT has *massive* prior art [2]; the phone tract has no true focal Fourier plane, so in practice the transform is partly computed on the CPU and the optics illustrate it.

Grover / Deutsch–Jozsa / CNOT / teleportation. Sequences of such linear maps plus an $\arg\max$ readout. They reproduce the textbook circuit *algebra* on small inputs but provide no quantum speed-up: the classical encoding pays at least linearly for what a quantum register would hold in superposition.

"Measurement" is $\arg\max$ over the captured intensities — a deterministic classical selection, not a projective measurement with Born-rule statistics over single quanta.

2.3 Wave-optical encoding and the Talbot carpet

The natural diffraction of a periodic display pattern is the substrate for these transforms. A grating of period a re-images itself at the Talbot distance $z_T = 2a^2/\lambda$ and produces fractional self-images at $z_T/2$, $z_T/4$, ... (Figure 1). This Fresnel self-imaging is what lets a displayed basis pattern survive the air gap and arrive structured at the camera; it also bounds the usable mode count. The carpet is genuine wave optics — and equally genuinely classical.

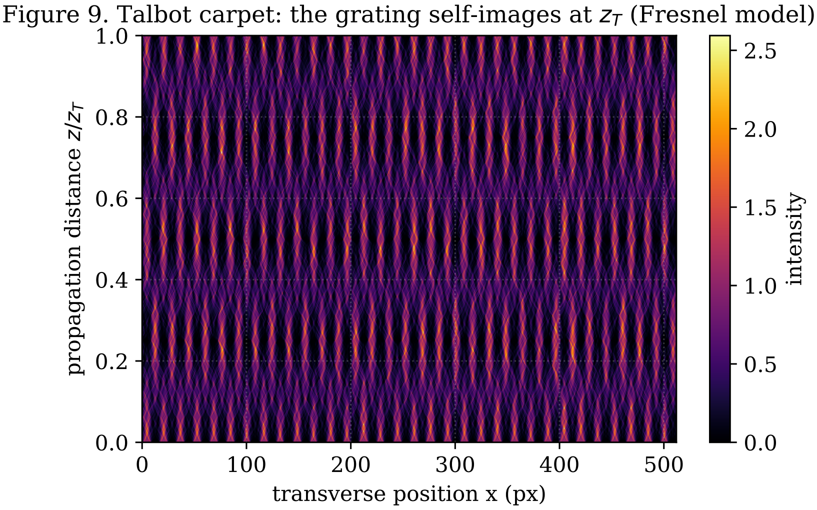

Figure 1: Talbot carpet (Fresnel self-imaging of a periodic display pattern). Self-images recur at $z_T = 2a^2/\lambda$ and its rational fractions. This classical near-field diffraction is the wave-optical substrate on which the "gate" patterns are encoded and read back — and it is fully classical.

2.4 Why this is not quantum computing — the explicit non-claims

We make the following non-claims prominently, because they are the point of the paper:

1. **No single photons.** The OLED emits $\sim 10^{10}$ photons per frame per region; every measurement is in the bright, classical regime.
2. **No entanglement.** Spatial modes of one classical beam can mimic two-qubit algebra (Spreeuw), but they are separable classical degrees of freedom; there is no nonlocal correlation and no 2^n-scaling tensor-product state space.
3. **No Bell-inequality violation.** A real CHSH test [3] requires space-like-separated measurements on entangled quanta and the violation of $|S| \leq 2$. Our correlator is local and classical and cannot violate any Bell inequality.
4. **No no-cloning / no QKD security.** The BB84-labelled demo copies classical patterns freely; with no single-photon non-cloning it offers **no** information-theoretic key security. It is a classical pattern-exchange illustration, not QKD.

3 A classical lock-in correlator at the display refresh (not a Bell test)

The one element of the "quantum" cluster worth keeping is the phase-sensitive correlator (repository B11; stages `chsh`, `aspect`, `hom`). The display refresh (120 Hz) provides a stable reference modulation. Patterns are emitted with a known phase/frequency; the captured brightness of each region is demodulated into in-phase and quadrature components,

$$I_{\cos} = \frac{2}{N}\sum_n y_n \cos(\omega t_n), \qquad I_{\sin} = \frac{2}{N}\sum_n y_n \sin(\omega t_n), \tag{2}$$

with $\omega = 2\pi\,(120\,Hz)$, and the angular correlation between two regions is estimated from $\sqrt{I_{\cos}^2 + I_{\sin}^2}$ and its phase, averaged over many frames with a t-test. This is an ordinary **classical lock-in amplifier** on a camera stream — a useful, narrow technique for pulling weak periodic optical correlations out of noise.

We stress what it is **not**. Although historically named after CHSH/Aspect, the stage performs no Bell test. The CHSH bound $|S| \leq 2$ for local hidden-variable theories, and its quantum violation up to $2\sqrt{2}$, concern measurements on *entangled single quanta* with space-like separation [3]. Our correlator measures classical intensities of one bright beam; any "S" it computes is a classical correlation function bounded trivially by classical statistics. **The Bell/quantum interpretation is removed; the lock-in correlator is retained as a classical method.**

4 The falsification protocol: the real contribution

The central risk in any physical-computing claim — acute when the labels say "quantum" — is that the apparatus is theatre and the CPU does the work. The Svetoch answer is a **mirror-less control**: run the identical software pipeline with the screen facing an open room, removing the optical return path. If a metric survives, it was a software artifact; if it collapses, the optics were computing. Applied to the gate emulations, the metrics collapse (Figure 2, Table 1).

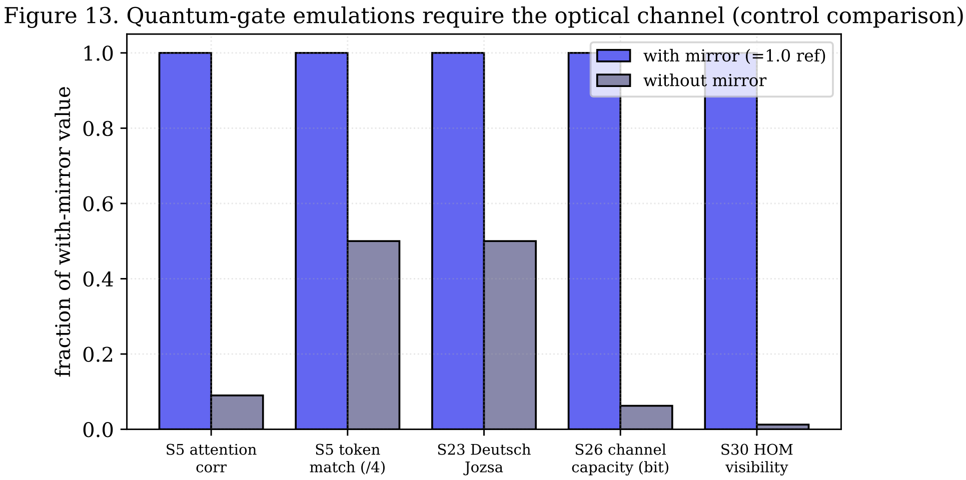

Figure 2: Quantum-gate emulation metrics, with mirror vs. without mirror (each bar is the without-mirror value as a fraction of the with-mirror value). Every emulation that needs the optical channel falls to the noise/coin-flip floor when the mirror is removed.

The Deutsch–Jozsa result is the clearest tell: with the mirror the procedure returns the correct constant/balanced verdict; without it the verdict is 50% — a fair coin. A genuine software computation would be unaffected by removing a mirror. The HOM-labelled "visibility" falls 80× to essentially zero, and the channel capacity drops from 16 bits to 1 bit (a single on/off level surviving via stray light). These collapses prove nothing quantum — only that there *is* a real classical optical channel and that the emulations use it. That distinction, demonstrated rather than asserted, is the contribution.

The same control underlies Paper I (Figure 3): removing the mirror drops the dynamic range $1.42 \to 1.000047$ and weakens white–black contrast by thousands of times across all stages.

Table 1: Mirror vs. mirror-less control for the gate emulations (documented reference values). Without the optical channel the emulations reduce to noise or coin-flips.

Metric (stage)	With mirror	Without mirror	Interpretation
S5 attention correlation	1.00	0.09	No optical feedback
S5 token match	4/4	2/4	Random guessing
S2 dot-product correlation	0.976	-0.41	Anti-correlated noise
S23 Deutsch–Jozsa	correct	50%	Coin flip
S26 channel capacity	16 bit	1 bit	Channel gone
S30 HOM visibility	0.056	0.0007	80× lower
Dynamic range	1.42	1.000047	No optical channel

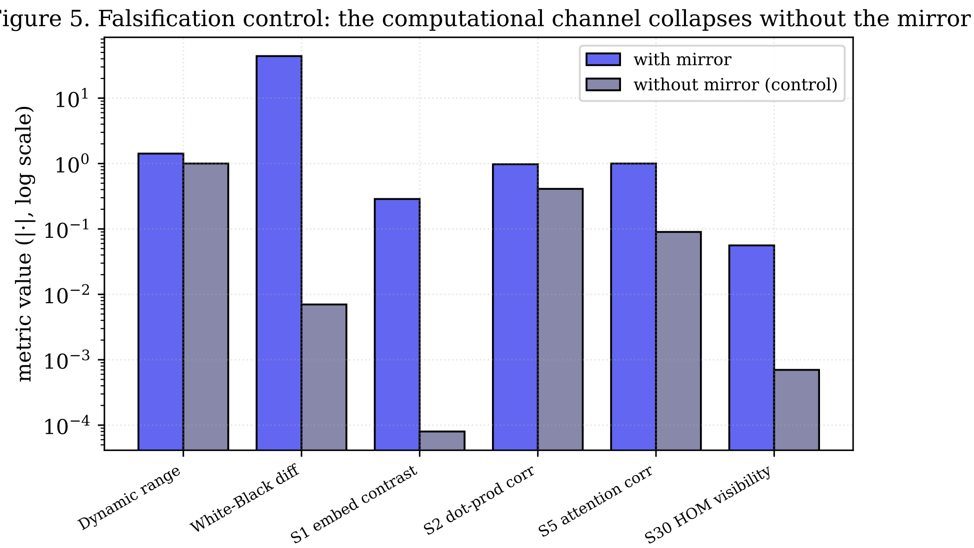

Figure 3: General falsification control (log scale): for every computational metric the with-mirror value towers over the mirror-less value. The optical channel, not the CPU, carries the signal.

5 What survives without the mirror, and why

We report the survivors honestly, because they bound the contribution.

Shot-noise random numbers are real. CMOS dark-frame and shot-noise entropy do not need the mirror; they are a true physical entropy source and pass standard randomness checks [5]. We claim no novelty: extracting randomness from photosensor shot noise has substantial prior art, and we present it only as the one component that is physically real *and* mirror-independent.

Monotone stages limp via scattered light. A weak scattered-light path ($\sim$0.3% of OLED light reaches the camera through the room and internal cross-talk) lets soft-threshold, monotone stages partially track their inputs without the mirror — which is exactly why we do not count them as evidence of optical computation. A monotone activation correlating with a monotone input proves only monotonicity, not channel structure.

Spatially resolved stages fail. Everything requiring real spatial resolution — embeddings, dot products, the gate-emulation interference patterns, the lock-in correlations — collapses without the mirror (Section 4). That selective failure is the diagnostic we rely on: real optical effects need the optics; CPU artifacts do not.

6 Methods

Hardware: Xiaomi 12 Lite (6.55" AMOLED, 2400 × 1080, 120 Hz, pitch 63.2 µm; 32 MP front camera, min. focus ≈10 cm); flat mirror ∼10 ×10 cm; gap $d \approx$ 3–5 cm in a dim room. Control runs are identical except the mirror is removed and the screen faces an open room. **Software:** a dependency-free Python HTTPS server relays Start/Stop and stores runs as JSON; experiments are browser ES modules. For each gate stage the phone renders the basis/interference pattern, captures frames with `getUserMedia`, white-normalizes, γ-corrects, computes the stage metric (interference contrast, arg max verdict, lock-in $I_{\cos}/I_{\sin}$, HOM visibility, capacity), and posts results; the lock-in demodulation uses the 120 Hz refresh as reference and aggregates many frames with a t-test. **Figures:** produced by `papers/scripts/make_figures.py`; real-data figures use a reference run (2026-06-06); the Talbot carpet (Figure 1) is a wave-optics model, labelled as such.

7 Discussion

The value of these experiments is as a **teaching and analogue platform**, not as a quantum computer. They let a student build the algebra of Hadamard, QFT, Grover and Deutsch–Jozsa from brightness patterns and interference on hardware they already own, and — crucially — test whether the apparatus does anything physical by removing the mirror. The discipline of the falsification control transfers well beyond this project: any claim that "a physical substrate computes X" should come with a null configuration in which the substrate is removed and the metric is shown to collapse.

Explicit non-claims (restated). No single photons; no entanglement; no Bell-inequality violation; no no-cloning and hence no QKD security; no quantum speed-up. The "QFT-by-diffraction" has heavy Fourier-optics prior art and is not a true focal-plane transform on this device. The shot-noise TRNG is physically real but has prior art. The CHSH/Aspect-labelled stage is a classical lock-in correlator, not a Bell test, and its quantum interpretation is withdrawn.

Limitations. The classical encoding pays at least linearly in the number of emulated qubits, so the analogue does not scale; ambient light, vibration and mirror quality degrade contrast; the pass thresholds are device-specific; cross-device reproducibility is an open validation shared with the rest of the series.

8 Conclusion

A smartphone and a mirror can faithfully **emulate the linear algebra of quantum gates with classical wave optics** — and a mirror-removal control proves these are classical optical effects, not CPU theatre and not quantum computation. We have set out exactly what is and is not being computed, retained the one defensible measurement primitive (a classical lock-in correlator) on honest terms, noted the real-but-prior-art shot-noise TRNG, and given the falsification protocol whose collapse numbers (HOM 0.056 → 0.0007, Deutsch–Jozsa → 50%, capacity 16 → 1 bit) are the project's honesty test. We publish this framing openly so the record is unambiguous: an honest classical analogue, not a quantum claim.

Data and code. github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vm (Apache-2.0); reference-run data under `examples/` and `papers/scripts/`.

Priority note. Released as a defensive publication to establish authorship and date of disclosure — in particular the honest classical-analogue framing and the falsification protocol; the author has elected not to seek patent protection.

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Part of the *Svetoch* series (defensive publication, not patented). Project and code: github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vmf. Released for the public record to establish authorship and priority.